

Source: Local Subject Media

24CS206 MACHINE LEARNING

Hours Per Week :	L	T	P	SL	C
	2	2	2	2	4

PREREQUISITE KNOWLEDGE: Probability & Linear Algebra, Python language.

COURSE DESCRIPTION AND OBJECTIVES:

This course introduces fundamental machine learning concepts, including supervised and unsupervised learning (clustering and dimensionality reduction). It helps students understand key challenges and select appropriate model complexity. Through practical case studies in areas such as computer vision, medical imaging, audio, and text, students gain real-world exposure. The lab component focuses on hands-on implementation using Python libraries like Scikit-learn and Matplotlib.

MODULE-1

UNIT-1 **6L+6T+6P+6SL=24 Hours**

INTRODUCTION

What is machine learning? Machine learning applications; Types of Learning: Supervised learning, unsupervised learning, and reinforcement learning. (T1,pp.1-9)

Model Training Essentials: Bias–Variance Trade-off (T2,pp. 67–69), Hypothesis Testing, Re-sampling methods: Sub sampling and Upsampling, SMOTE; Cross Validation (validation set, Leave-One-Cut (LOO), k-fold strategies),(R2,pp.198-215); Evaluation Metrics-Error functions, Confusion Matrix, Accuracy, Precision and Recall, F1 Score (R3,pp.86-89).

Regression Analysis: Linear Regression - Simple and Multiple Linear Regression (R2, pp.63-92), Polynomial Regression (R1, pp.290), Logistic Regression (R1,pp-119-127). Model Shrinkage-Ridge, and LASSO regression (R2, pp.237-241).

UNIT-2 **6L+6T+6P+6SL=24 Hours**

FEATURE SELECTION

Feature Selection Strategies: Types of feature selection Filter method: Chi-Square test, Wrapper method: forward/backward selection algorithms, Embedded method: Elastic net.(R3,pp.38-47)

Dimensionality Reduction: Matrix factorization(T1, pp.561-569), Principal components analysis, Linear discriminant analysis (T3, pp.207-226).

PRACTICES:

Exercise 1: ML Pipeline Construction

- Dataset: Any tabular dataset (e.g., student performance / housing)

Tasks:

- Identify problem type (classification/regression)
- Perform preprocessing (missing values, encoding, scaling)
- Split into train–test sets

Analysis Task: Justify chosen learning type (supervised/unsupervised)

Exercise 2: Cross-Validation Techniques

Tasks:

- Implement:
- Simple validation
- k-fold (k=5,10)
- Leave-One-Out (LOO)
- Compare accuracy

Analysis Task: Trade-off between computational cost and reliability

Exercise 3: Handling Imbalanced Dataset

• Dataset: Fraud detection / medical dataset

Tasks:

• Apply:

• Undersampling

• SMOTE

Analysis Task: Compare impact on recall and precision

Exercise 4: Linear vs Polynomial Regression

Tasks:

• Fit linear regression

• Fit polynomial regression (degree 2,3)

• Compare RMSE

• Train:

• Simple model (underfitting)

• Complex model (overfitting)

• Plot training vs validation error

Analysis Task: Identify overfitting, and bias vs variance issues

Exercise 5: Multiple Linear Regression

Tasks:

• Train model with multiple features

• Interpret coefficients

Analysis Task: Which features are most influential?

Exercise 6: Logistic Regression

Tasks:

• Binary classification using logistic regression

• Multi-class using multinomial regression

• Train model

• Compute:

• Accuracy, Precision, Recall, F1-score

• Plot confusion matrix

Analysis Task: Compare decision boundaries

Exercise 7: Regularization (Ridge vs LASSO)

Tasks:

• Train:

• Ridge regression

• LASSO regression

• Compare coefficients

Analysis Task:

• Which features are eliminated by LASSO?

Exercise 8: Feature selection Techniques

Tasks:

• Apply Chi-square test

• Rank features

• Implement:

• Forward selection

• Backward elimination

• Train Elastic Net model

• Analyze feature importance

Analysis Task: Select top-k features and justify

• Compare with Ridge & LASSO

SKILLS:

- ✓ *Statistical data analysis.*
- ✓ *Classify / Cluster data.*
- ✓ *Tool usage for developing ML applications.*

Exercise 9: LDA vs PCA

Tasks:

• Apply PCA and LDA on same dataset

• Visualize projections

Analysis Task:

• Which gives better class separation and why?

Exercise 10: End-to-End Model Optimization

• Dataset: Real-world dataset

Tasks:

• Perform:

• Feature selection

• Dimensionality reduction

• Model training (Regression/Classification)

• Evaluate using cross-validation

Analysis Task:

• Which combination gives best performance?

MODULE-2

UNIT-1 **6L+6T+6P+6SL=24 Hours**

CLASSIFICATION: Classification Types: Binary, Multi-class, and Multi-label Classification, (T2 pp.254-257) Decision tree induction - attribute selection measures; Tree pruning; CART (Classification and Regression Trees) (T2 pp.216), k-NN (k-Nearest Neighbor classifier) (T3 pp 64-66) and Fuzzy k-NN, (R3, pp.219).

UNIT-2 **6L+6T+6P+6SL=24 Hours**

ENSEMBLE METHODS: Voting: Soft Voting, Hard Voting, Bagging: Random Forest Trees, Boosting: Gradient Boosting, XGBoost, Stacking (T3, pp.183-202).

UNIT-3 **6L+6T+6P+6SL=24 Hours**

Clustering Techniques: Hierarchical methods; Agglomerative and divisive method, Fuzzy C-Means Clustering (R4, pp.578-581), Spectral Clustering (R5. Pp.396-413), Cluster Evaluation Methods (R4, pp.423-430).

PRACTICES:

Exercise 1: Binary Classification Pipeline

• Dataset: Breast Cancer / Spam dataset

Tasks:

• Perform preprocessing (missing values, normalization)

• Train a Logistic Regression / Decision Tree classifier

• Evaluate using Accuracy, Precision, Recall, F1-score

• Plot confusion matrix

Analysis Task: Interpret false positives vs false negatives

Exercise 2: Multi-class Classification

• Dataset: Iris / Digits

Tasks:

• Implement multi-class classification using k-NN and Fuzzy k-NN (assign membership values)

• Compare classification result with k = 3, 5, 7

• Plot decision boundaries

Analysis Task: Explain effect of k on bias-variance, When does fuzzy approach outperform k-NN?

Exercise 3: CART Implementation

Tasks:

• Train Decision Tree using Gini Index

• Visualize tree structure

• Apply pruning

Analysis Task: Compare overfitting before/after pruning

Exercise 4: Voting Classifier

Tasks:

• Train 3 models (Logistic Regression, k-NN, Decision Tree)

• Combine using:

• Hard Voting

• Soft Voting

Analysis Task: Which voting performs better and why?

Exercise 5: Random Forest

Tasks:

• Train a Decision Tree and Random Forest

• Compare accuracy and variance

Analysis Task: Explain why Random Forest reduces overfitting

Exercise 6: Boosting Techniques

Tasks:

• Train Gradient Boosting / XGBoost

Analysis Task:

• Compare performance across models

• Identify cases where boosting fails

Exercise 7: Stacking

Tasks:

• Use 2 base learners + 1 meta-learner

• Compare with individual models

Analysis Task: Explain role of meta-learner

Exercise 8: Fuzzy C-Means Clustering

Tasks:

• Implement FCM algorithm

• Visualize membership values

• Compute:

• Silhouette Score, Davies-Bouldin Index

Analysis Task: Compare with K-Means clustering

Exercise 9: End-to-End ML Comparison

• Dataset: Any real-world dataset (e.g., healthcare/customer data)

Tasks:

• Apply:

• Classification (k-NN, CART)

• Ensemble (Random Forest, XGBoost)

• Clustering (FCM / Spectral)

• Compare results

Analysis Task:

• When to use classification vs clustering?

COURSE OUTCOMES:

CO No.	Course Outcome	Bloom's Level	Module No.	Mapping with POs
C1	Apply a wide variety of learning algorithms such as Probabilistic, Discriminative and Generative algorithms for a given application.	Apply	Module 1 Unit 1, Module 2 Unit 1 & 2	PO1, PO2, PO3
C2	Design an end-to-end Machine-learning model to realize solutions for real-world problems.	Create	All Modules	PO1, PO2, PO3, PO5

CO No.	Course Outcome	Bloom's Level	Module No.	Mapping with POs
C3	Implement various machine learning models using advanced ML tools.	Apply	Module 1 Unit 1 & 2, Module 2 Unit 1, 2, 3	PO2, PO3, PO4
C4	Analyze and evaluate the performance of various machine learning model approaches on different kinds of data.	Analyze& Evaluate	Module 1 Unit 1, Module 2 Unit 3	PO1, PO2, PO4
C5	Develop expertise in ensemble methods and their application in improving predictive performance.	Apply	Module 2 Unit 2	PO2, PO3, PO4

MAPPING OF SUSTAINABLE DEVELOPMENT GOALS (SDGS) & IKS:





TEXT BOOKS:

REFERENCE BOOKS:



Source: Local Subject Media

24CS206 WEB TECHNOLOGIES

Hours Per Week :	L	T	P	SL	C
	0	2	2	0	2

Prerequisite: Object-Oriented Programming (Java or Python)

Course Description:

This course bridges the gap between traditional full-stack web development and modern artificial intelligence. It offers comprehensive training in building dynamic web applications using HTML5, CSS3, modern JavaScript (ES6+), and Node.js, while seamlessly integrating AI capabilities. Students will learn to leverage AI coding assistants, implement prompt engineering, integrate Large Language Model (LLM) APIs, and utilize vector databases. The ultimate objective is to empower students to design, develop, and deploy intelligent, agentic web applications that can understand, process, and generate dynamic content.

MODULE-1: MODERN WEB FOUNDATIONS & CLIENT-SIDE INTELLIGENCE

Flow Logic: We start by building static pages, add logic with JS, introduce modern frontend frameworks, and then inject AI directly into the user's browser.

UNIT-1 **0L+6T+6P+0SL=12 Hours**

CORE WEB TECHNOLOGIES & AI-ASSISTED DEVELOPMENT

Modern HTML5 & CSS3: Semantic HTML, Canvas, Audio and Video Integration, CSS3 Grid, Flexbox, Responsive Design, CSS Variables. JavaScript (ES6+): DOM Manipulation, Arrow Functions, Promises, Async/Await, Closures, Fetch API, Event Handling, Regular Expression Validation. Code Debugging, Boilerplate Code Generation. [TB2: Chapters 1-9 & 10-15], [RB1: Chapters 3 & 11], [WL1: JavaScript Guide], [WL3: AI Assistant Setup Docs]

UNIT-2 **0L+6T+6P+0SL=12 Hours**

CLIENT-SIDE FRAMEWORKS & BROWSER AI

Frontend Frameworks: Component Architecture, State Management, Props and Data Binding, Routing (Angular/React). Intelligent UI/UX: Conversational Interfaces, Chat UI, Adaptive Dynamic Forms. Browser-Based AI: Web Speech API, Speech-to-Text, Text-to-Speech, Browser-based Pre-trained Models, TensorFlow.js, Image Classification, Client-side Sentiment Analysis. [TB1: Chapters 17-20], [TB3: Chapters 2, 4 & 8], [RB2: Chapter 4], [WL2: Component Architecture Docs]

PRACTICE:

Design a responsive, multi-page personal portfolio using HTML5 and CSS3. Utilize an AI coding assistant to generate the initial scaffolding, and manually refine the code for optimal layout.

Develop an interactive catalog page for books or products using JavaScript. Implement client-side filtering and sorting functionalities that update the DOM without requiring page reloads.

Design a comprehensive student registration form. Validate the inputs using standard JavaScript regular expressions and integrate a lightweight client-side AI model to detect inappropriate language in the text areas.

Create a voice-controlled web page featuring a color palette. Utilize the Web Speech API so that when a user speaks a command (e.g., "Make background blue"), the web page background updates accordingly.

Develop a web application that captures an image via the user's webcam and utilizes pre-trained TensorFlow.js models in the browser to identify and label the primary object in the frame.

MODULE-2 SERVER-SIDE DATA & INTELLIGENT WEB AGENTS

Flow Logic: We build a secure backend server first, then connect that server to LLMs and Vector DBs to give it "brains," and finally upgrade the application to act as an autonomous agent before deploying.

UNIT-1 **0L+6T+6P+0SL=12 Hours**

SECURE BACKEND DEVELOPMENT & DATABASES

Node.js & Express: Node.js Server Setup, Express Routing, Middleware, RESTful APIs, CORS. Authentication & Sessions: JSON Web Tokens (JWT), Secure Login/Signup Flows, Protected Routes, Session Management. Database Integration: NoSQL Fundamentals, MongoDB, CRUD Operations, Mongoose ODM. [TB1: Chapters 5, 6, 8, 13, 14 & 26], [WL1: HTTP Authentication Guide]

UNIT-2 **0L+6T+6P+0SL=12 Hours**

LLM INTEGRATION & CONTEXTUAL MEMORY

API Integration: OpenAI API, Google Gemini API, Secure API Key Management, Environment Variables. Prompt Engineering: System Prompts, Few-shot Prompting, Context Window Management, AI Hallucination Mitigation. Vector Databases & RAG: Data Embeddings, Vector Database Setup (Pinecone/Weaviate/MongoDB Atlas Vector Search), Semantic Data Storage. [RB2: Chapters 2 & 5], [WL3: Node.js Quickstart], [WL4: Q&A; with RAG], [WL5: Understanding Embeddings]

UNIT-3 **0L+6T+6P+0SL=12 Hours**

BUILDING AGENTIC TOOLS & DEPLOYMENT

Agentic Web Workflows: Autonomous AI Tasks, Tool Calling, Automated Web Scraping, Content Summarization Pipelines. Deployment & Security: Application Hosting (Vercel/AWS EC2), Secure Environment Configurations, Prompt Injection Security, Endpoint Rate Limiting. [TB1: Chapter 32], [WL4: Agents and Tools], [WL2: Deployment Guides]

PRACTICE:

Design a RESTful API utilizing a Node.js and Express backend connected to a MongoDB database to perform complete CRUD operations for a library management system.

Implement a secure authentication system featuring login and signup flows using JSON Web Tokens (JWT) to protect specific application routes and manage user sessions.

Develop a custom frontend chatbot interface connected to a Node.js backend. Securely format user inputs, make calls to an LLM API, and stream the generated AI responses back to the client interface.

Design a semantic search engine by embedding and storing a dataset of university course descriptions in a vector database. Allow users to search for courses using natural language queries rather than exact keyword matches.

Develop an intelligent web agent (Capstone Application) where a user can upload a document or provide a URL. Program the backend to extract the text, utilize an LLM to generate a summarized report, translate it into a specified language, and render the final results on a secure user dashboard.

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Module No.	Mapping with POs
CO1	Understand the foundational concepts of modern web technologies, including semantic HTML5, CSS3, JavaScript (ES6+), and AI-assisted development tools.	Understand	M-I U-I	PO1, PO2, PO5
CO2	Describe the component architecture of modern front-end frameworks and the mechanisms of browser-based AI APIs like Web Speech and TensorFlow.js.	Understand	M-I U-II	PO1, PO2
CO3	Apply Node.js, Express, and NoSQL databases to build and secure server-side RESTful APIs with JWT authentication.	Apply	M-II U-I	PO1, PO2, PO3, PO5

SKILLS:

- ✓ *Develop functional web and mobile applications using modern AI coding assistants.*
- ✓ *Design effective prompt engineering workflows for architecture and UI generation.*
- ✓ *Integrate frontend components with backend databases, APIs, and cloud services.*
- ✓ *Debug, test, and safely deploy AI-generated application code.*

CO No.	Course Outcomes	Blooms Level	Module No.	Mapping with POs
CO4	Analyze prompt engineering strategies and Retrieval-Augmented Generation (RAG) architectures for integrating Large Language Models and vector databases.	Analyze	M-II U-II	PO1, PO2, PO3, PO4
CO5	Design and deploy comprehensive, multi-tier intelligent web applications that execute autonomous agentic workflows while addressing endpoint security.	Create	M-II U-III	PO2, PO3, PO5, PO8

TEXTBOOKS (T):

Brad Dayley, Node.js, MongoDB and Angular Web Development, 2nd Edition, Addison-Wesley Professional, 2017.

Jon Duckett, HTML and CSS: Design and Build Websites (Part 1) & JavaScript and JQuery: Interactive Front-End Web Development (Part 2), Wiley.

Gant Laborde, Learning TensorFlow.js: Powerful Machine Learning in JavaScript, O'Reilly Media, 2021.

REFERENCES (R):

Cay S. Horstmann, Modern JavaScript for the Impatient, Addison-Wesley Professional, 2020.

Olivier Caelen, Marie-Alice Blete, Developing Apps with GPT-4 and ChatGPT: Build Intelligent Chatbots, Content Generators, and More, O'Reilly Media, 2023.

Web Links & Official Documentation (W):

MDN Web Docs (Mozilla Developer Network) - Standard for HTML, CSS, JS, and Web APIs.

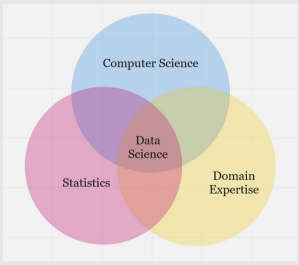
(https://developer.mozilla.org)

React Official Documentation. (https://react.dev)

Google Gemini API Documentation / OpenAI API Documentation. (https://ai.google.dev/docs | https://platform.openai.com/docs)

LangChain.js Documentation - Standard for RAG and Agentic workflows. (https://js.langchain.com/docs)

Pinecone / MongoDB Atlas Vector Search Docs. (https://docs.pinecone.io | https://www.mongodb.com/docs/atlas/atlas-vector-search)



Source: Local Subject Media

25CY206 - STATISTICAL FOUNDATION FOR COMPUTING AND DATA SCIENCE

Hours Per Week :	L	T	P	SL	C
	3	0	2	3	4

PREREQUISITE KNOWLEDGE: Basic knowledge in Mathematics and Statistics.

COURSE DESCRIPTION:

This course introduces students to probability theory, statistical inference, and computational statistical techniques essential for engineering, Artificial Intelligence, Machine Learning, Data Science, and scientific computing applications. It provides a rigorous foundation in random variables, probability distributions, sampling theory, estimation, hypothesis testing, regression analysis, and multivariate statistical methods. The course emphasizes statistical reasoning under uncertainty and data-driven decision-making. It also integrates computational tools and AI-based platforms for visualization, simulation, and analysis of statistical models. The course enables students to develop analytical thinking, mathematical modeling skills, and computational proficiency required for modern data-centric engineering systems and intelligent computing applications.

COURSE OBJECTIVES:

Understand fundamental probability theory and random variables.

Apply probability distributions in computing and data science problems.

Develop statistical inference and estimation techniques.

Perform hypothesis testing for engineering decision-making.

Analyze relationships using regression models.

Introduce PCA and time series forecasting for AI systems.

Strengthen computational and AI-assisted statistical skills.

MODULE-1

18L+0T+12P+18SL=48Hours

UNIT 1

FUNDAMENTALS OF PROBABILITY

Introduction to probability and its applications in computing, Probability axioms, Conditional probability and independence, Bayes' theorem and its applications. Prior probability and posterior probability; Bayesian inference; Maximum A Posteriori (MAP) estimation.

UNIT 2

RANDOM VARIABLES & PROBABILITY DISTRIBUTIONS

Random variables: Distribution function, discrete random variable, probability mass function, continuous random variable, probability density function, expectation, variance and their properties

Discrete probability distributions: Uniform, Binomial, Poisson. Continuous probability distributions: Uniform, Normal, Exponential, Weibull, Cauchy.

PRACTICES:

Practice No.	Practice	AI Tools / Platforms
1	Explore axioms of probability by formulating real-life event spaces (coin, dice, card experiments) and verifying probability rules such as non-negativity, normalization, and additivity	ChatGPT, Claude AI

Practice No.	Practice	AI Tools / Platforms
2	Visualize conditional probability using Venn diagrams and real-world scenarios such as disease testing or spam filtering systems	GeoGebra AI, Desmos
3	Analyze Bayes' theorem in decision-making problems such as medical diagnosis and email classification and interpret posterior probability updates	Microsoft Copilot, Perplexity AI
4	Compare discrete and continuous random variables by examining PMF vs PDF representations and interpreting their graphical behaviour	Wolfram Alpha, Symbolab
5	Investigate Binomial and Poisson distributions by modeling real-world systems such as defect detection and arrival processes	ChatGPT, Gemini AI
6	Compute expectation, variance, and higher moments for given datasets and interpret their statistical significance in engineering data	Claude AI, ChatGPT
7	Construct probability trees for multi-stage experiments such as machine failure analysis or multi-step decision systems	Perplexity AI, Gemini AI
8	Simulate random variables from different probability distributions and compare theoretical vs empirical results	Wolfram Alpha, ChatGPT
9	Analyze Normal and Exponential distributions using real-world datasets.	ChatGPT, Gemini AI
10	estimate distribution parameters, visualize probability distributions, and interpret the results for Normal and Exponential distributio	ChatGPT, Gemini AI

SKILLS:

- ✓ *Develop functional web and mobile applications using modern AI coding assistants.*
- ✓ *Design effective prompt engineering workflows for architecture and UI generation.*
- ✓ *Integrate frontend components with backend databases, APIs, and cloud services.*
- ✓ *Debug, test, and safely deploy AI-generated application code.*

MODULE-2

27L+0T+18P+27SL=72Hours

UNIT 1

ESTIMATION AND TESTING OF HYPOTHESIS:

Population and sample, sampling distributions, point estimation and properties of estimators, introduction to Maximum Likelihood Estimation (MLE), and confidence intervals for population parameters. Hypothesis testing: Null and alternative hypotheses, level of significance, p-value, critical value, Type I and Type II errors, Single sample and two sample tests for mean's and variance's and chi-square test-Independence attributes. A/B Testing and its applications in data-driven decision-making and AI-enabled systems.

UNIT 2

CORRELATION & REGRESSION ANALYSIS

Correlation analysis: scatter plots, Pearson's correlation coefficient, and interpretation. Simple linear regression model and assumptions, regression coefficients, fitting and interpretation of the regression line, coefficient of determination (R^2), prediction, and residual analysis. Introduction to data transformations (log and square root) and Introduction to logistic regression for binary classification.

UNIT 3

ADVANCED STATISTICAL METHODS

Introduction to Multivariate Analysis (MVA), multivariate data, covariance and correlation matrices, Principal Component Analysis (PCA), dimensionality reduction, interpretation and applications of principal components. Introduction to Time Series Analysis, time series data, components of a time series, trend estimation using the least squares method, moving averages, and simple exponential smoothing for forecasting.

PRACTICES:

Practice No.	Practice	AI Tools / Platforms
1	Analyze sampling distributions and estimate population parameters	ChatGPT, Perplexity AI
2	Explore Maximum Likelihood Estimation by interpreting likelihood functions for statistical models	Wolfram Alpha, Claude AI
3	Construct and interpret confidence intervals for population parameters in engineering datasets	Microsoft Copilot, Gemini AI
4	Perform hypothesis testing (Z-test, t-test, chi-square test) and interpret statistical significance	ChatGPT, Symbolab
5	Compare parametric tests (Z, t, F, chi-square) based on assumptions and applicability	Wolfram Alpha, ChatGPT
6	Visualize simple linear regression models including regression line and residual distribution	GeoGebra AI, Desmos
7	Analyze logistic regression for binary classification problems such as prediction tasks	Perplexity AI, Gemini AI
8	Apply PCA for dimensionality reduction and interpret principal components in feature space	NotebookLM, ChatGPT
9	Investigate time series data patterns including trend and seasonality in real datasets	Microsoft Copilot, Claude AI
10	Validate statistical inference results and compare theoretical vs computed conclusions	Wolfram Alpha, ChatGPT

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to

CO No.	Course Outcomes	Blooms Level	Mapping with POs
1.	Understand the fundamental concepts of probability, random variables, and probability distributions and apply them in computing applications.	Apply	1,2,12
2.	Analyze different probability distributions, their properties, and significance in real-world applications, including computing and machine learning.	Analyze	1, 2, 4
3.	Perform statistical inference using estimation and hypothesis testing techniques, and interpret the results effectively.	Evaluate	2,3,5

CO No.	Course Outcomes	Blooms Level	Mapping with POs
4.	Apply regression analysis techniques, including linear and logistic regression, and interpret results.	Apply	2,4,6
5.	Evaluate the application of Principal Component Analysis (PCA) and Time Series Analysis for data-driven decision-making.	Evaluate	3,5,11

MAPPING OF SUSTAINABLE DEVELOPMENT GOALS (SDGS) & IKS:





TEXT BOOKS:

- Sheldon M. Ross, Introduction to Probability and Statistics for Engineers and Scientists, 5th Edition, Academic Press, 2019.
- Jay L. Devore, Probability and Statistics for Engineering and the Sciences, 9th Edition, Cengage Learning, 2015.

REFERENCE BOOKS:

- Morris H. DeGroot, Mark J. Schervish, Probability and Statistics, 4th Edition, Pearson, 2011.
- Casella, George, Roger L. Berger, Statistical Inference, 2nd Edition, Cengage Learning, 2002.
- Christopher M. Bishop, Pattern Recognition and Machine Learning, Springer, 2006.
- SKILLS
- Statistical reasoning and probabilistic modeling
- Analytical thinking for engineering problem solving
- Mathematical modeling of real-world systems
- Data-driven decision-making ability
- Machine learning and AI-assisted statistical analysis
- Regression and predictive modeling techniques
- Dimensionality reduction and feature engineering
- Time series forecasting and trend analysis
- Python-based scientific computing skills
- Data visualization and interpretation skills
- Image source: https://www.google.com/search?q=PROBABILITY+%26+STATISTICS+normal+graph+IMAGE
- Image file name: PROBABILITY & STATISTICS